

Second Report Chapter (Team 9)

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1. Functional Design

Core Principles

Functional Design defines **what** a system must do before establishing physical solutions or technical mechanisms.

- **Solution Neutrality:** Functions are defined without committing prematurely to specific technologies or physical components.
- **System Boundaries & Flows:** The overall function maps system inputs into outputs across three primary categories:
 - **Energy (E):** Electrical, mechanical, or thermal energy transfers.
 - **Signal (S):** Control inputs, sensor data, or status feedback.
 - **Mass / Matter (M):** Materials, fluids, or workpieces passing through the system.
- **Function Categories:** Core operational actions used to structure sub-functions:
 - *Conversion* (changing input form to output form)
 - *Varying* (amplifying, stepping down, or modulating)
 - *Connecting/Disconnecting* (switching, coupling, or mixing)
 - *Channeling* (directing or transferring power, signals, or mass)
 - *Storing* (holding energy, signals, or mass)

2. Utility Value Analysis (*Nutzwertanalyse*)

Methodology

Utility Value Analysis is a quantitative multi-criteria decision-making framework used to evaluate, rank, and select competing design variants.

The 5-Step Evaluation Process

1. **Define Alternatives:** Identify concept variants generated during conceptual design.
2. **Select Evaluation Criteria:** Derive criteria directly from the system requirements list (e.g., performance, safety, manufacturability).
3. **Weight Criteria:** Assign relative weight factors (w_i) to criteria such that $\sum w_i = 1.0$ (or 100%).
4. **Score Alternatives:** Rate performance (v_i) of each concept against each criterion using a standardized rating scale.
5. **Calculate Overall Utility Value:** Compute total weighted utility:

$$\text{Utility Score} = \sum_{i=1}^n (v_i \cdot w_i)$$

Evaluation Scale Systems

- **Standard Use-Value Analysis (0 to 10 Scale):** Evaluates performance on an 11-point continuum where 0 represents *Useless*, 5 represents *Satisfactory*, and 10 represents *Ideal*.
- **VDI 2225 Guideline (0 to 4 Scale):** Standardizes rating across 5 discrete levels: 0 (*Unsatisfactory*), 1 (*Just tolerable*), 2 (*Adequate*), 3 (*Good*), and 4 (*Ideal / Very good*).

Methodological Limitations

- **Subjectivity:** Rating scores can be influenced by team bias or subjective interpretations.

- **False Precision:** Mathematical scoring creates an illusion of objective accuracy.
 - **Sensitivity:** Final rankings can shift significantly with minor changes to weighting distributions.
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3. Key Takeaways

- **Solution-neutral modeling** prevents premature design bias and clarifies fundamental system transformations.
- **Multi-criteria evaluation** transforms qualitative, multi-variable design trade-offs into structured, documentable decisions.